

NITISH DEVRANI

AI Engineer & Full-Stack Developer | Nuremberg, Germany

nitishdevrani@gmail.com | [linkedin.com/in/nitishdevrani](https://www.linkedin.com/in/nitishdevrani) | github.com/Nitishdevrani | nitishdevrani.github.io



Summary

AI engineer and full-stack developer with 6+ years shipping production web products — React and Next.js store-fronts serving millions of users at Petco, an airport-scale sampling platform for Adani One, live streaming for Babyflix — now doing applied AI research in Nuremberg on vision-language models, LLMs, custom AI agents and RAG systems. Builds end-to-end, from model experiments and evaluation to the APIs, interfaces and cloud deployment around them.

Skills

AI / ML	PyTorch, TensorFlow, scikit-learn, deep learning, LoRA fine-tuning, pruning & quantization, data engineering & augmentation
LLMs & Agents	LLM, VLM, RAG, MCP, LLM agents, prompt engineering, vector databases, custom chatbots, Claude, ChatGPT
Frontend	TypeScript, JavaScript, React, Next.js, Vue.js, Redux, React Native, PWA, HTML5, CSS3
Backend & Data	Node.js, NestJS, Python, REST APIs, WebSockets, Kafka, PHP, MongoDB, MySQL
Cloud & DevOps	AWS, Docker, Kubernetes, Terraform, GitLab CI/CD, Git
Design & CAD	Figma, UI/UX design, Onshape, AutoCAD, 3D printing; Raspberry Pi, ESP32 (embedded)
Languages	English (Fluent, C1), German (B2), Hindi (Native)

Experience

HiWi (Research Assistant)

Technische Universität Nürnberg (UTN), Nuremberg, DE

Sep 2025 – Present

React, TypeScript, Node.js, MongoDB, RAG, LLM

- Sole developer of the university's **Learning Companion**: architected and built frontend, REST backend, database and cloud deployment with CI/CD, and keep it running in production.
- Built lesson-assessment workflows across five Digitales Didaktisches Design dimensions — drafts, scored results, AI feedback and follow-up questions — with a dashboard for completed and in-progress assessments.
- Integrated AI group conversations with idea-generation, critique and synthesis perspectives on AcademicCloud-hosted LLMs, including follow-ups and saved chats.
- Developed **Multibot** and other RAG-powered tools, including a chatbot on a self-hosted transformer LLM that keeps data private with no third-party API.
- Implemented registration, email confirmation and role-based access for teachers, admins and super admins, plus admin tooling for AI instructions, health checks and logs.

Frontend Developer

Petco, Remote / India

Nov 2023 – Aug 2025

Next.js, TypeScript, SEO, Google Analytics

- Built, upgraded and maintained PDP, PLP, brand and homepage modules for a high-traffic US pet-retail store-front serving millions of users.
- Delivered SEO and UI/UX improvements that raised page ranking, discoverability and load performance, measured with Google Analytics.
- Worked in two-week Agile sprints with a distributed team across US and Indian time zones.

Software Engineer

Kellton (deployed at Adani Group), Gurugram, India

Mar 2022 – Nov 2023

React.js, Redux, PWA, REST, Docker, CI/CD

- Built the merchant-facing sampling frontend for **Adani One**, rolled out across India's major airports — barcode scan to sample claim to feedback — with real-time merchant analytics and a role-based admin panel.
- Delivered responsive, accessible React components, REST integrations and Progressive Web App capabilities.
- Automated form generation with a custom script, cutting repetitive setup for every new sampling campaign.

Sr. Associate Consultant

Oodles Technologies, Gurugram, India

Jan 2020 – Mar 2022

React.js, React Native, Vue.js, PHP, Drupal

- Led a team of 4 developers delivering **Babyflix**, a US pregnancy-tech live-streaming platform handling thousands of concurrent users, with private streaming rooms and event scheduling.
- Migrated the legacy PHP platform to React.js and drove architecture improvements for scalability, planning releases directly with stakeholders.
- Built apps for Apple TV; as Associate Consultant (Jan 2020 – Jan 2021) maintained the Drupal site and shipped new live-streaming features.

AI Research Projects

- **How Far Can You Compress CLIP?** (PyTorch, Wanda pruning, 4-bit quantization): 300+ experiments on CLIP ViT-B/32, ViT-B/16 and MobileCLIP-S1 comparing magnitude, activation-aware Wanda, structured FFN and EViT token pruning under six precision configurations and both prune/quantize orderings, with 30–70% sparsity sweeps and three-seed variance. MobileCLIP-S1 kept **99.62%** top-1 on ImageNette at 30% sparsity and 4-bit (FP32: 99.67%).
- **Memory-Augmented LLM Agent for MiniGrid** (Python, gpt-4.1-mini, BabyAI / MiniGrid): Co-authored paper (Feb 2026): an LLM navigation agent grounded in persistent spatial memory, a local symbolic map, subgoal prompting and action validation, evaluated on six BabyAI baseline tasks under partial observability.
- **Smart Bike Light Pro** (CNN, Raspberry Pi 5, ESP32, Onshape): Camera-to-headlight pipeline predicting off, dim or high beam with a lightweight residual CNN trained on 18,482 BDD100K images: **93.99%** policy-label accuracy, 20.4 ms inference on Raspberry Pi 5 in a 10 Hz control loop, in custom 3D-printed enclosures and bicycle mounts.
- **Cloud Specs — Multi-Cloud Advisor** (React, RAG, Qwen embeddings, vector DB): Platform comparing AWS, Azure and GCP compute instances with a weighted recommendation engine, a RAG pricing assistant (Arcana vector database, Qwen embeddings, Academic Cloud LLMs) and a conversational infrastructure shopping agent.
- **Cultural Bias in Large Language Models** (LLM evaluation, LLM-as-judge): Studied 330 responses to 30 culturally grounded prompts across five model families (GPT, Claude, Qwen, DeepSeek, Gemini), comparing 5 LLM judges with 9 human annotators; designed and ran the human evaluation.

Software Projects

- **Real-Time Data Monitoring Dashboard** (Next.js, Kafka, WebSockets, Recharts): Live dashboard for a Kafka/DuckDB query pipeline: query trends, query-type and cluster-utilisation charts, per-user billing, auto-dismissing threshold alerts with history, buffered predictions and WebSocket reconnect under frequent updates.
- **FITT, IIT Delhi — Website Revamp** (React, Next.js, Vue.js, TypeScript, Strapi CMS): Rebuilt the public site of IIT Delhi's Foundation for Innovation and Technology Transfer as a modern, responsive portal for startups, R&D partners and researchers.
- **3D Modeling & Printing** (Onshape, FDM printing): CAD-designed functional parts and models — household fixtures, hinges, the Smart Bike Light enclosures and mounts — optimised for print orientation and supports.

Education

M.Sc. in AI & Robotics, University of Technology Nuremberg (UTN), Germany	2024 – 2026
Master of Computer Applications (MCA), Gurugram University, India	2020 – 2022
Bachelor of Computer Applications (BCA), Maharishi Dayanand University, India	2017 – 2020

Papers & Reports

- **How Far Can You Compress CLIP? A Systematic Study of Pruning and Quantization for Edge Deployment** (Model compression): N. Devrani.
- **A Memory-Augmented LLM Agent for Open-Ended Navigation in MiniGrid** (LLM agents): V. Bytyqi, N. Devrani, P. Saxena, 2026.
- **Perception for Smart Bike Lights: Real-Time Embedded Bike Headlight Control Using a Single-Model CNN** (Embedded vision): S. Das, M. A. Siddiqui, N. Devrani — report and poster.
- **Cloud Specs: A Web App with a RAG-Powered Chatbot** (Cloud economics, RAG): N. Devrani.
- **Bias in AI: On Cultural Bias in Large Language Models** (AI ethics): N. Devrani, O. Nabail, A. Reyad.
- **A Perfectly Frictionless Companion Is a Perfectly Stunted One: On Sycophantic AI and the Eclipse of Moral Selfhood** (AI ethics, philosophy): N. Devrani.

Certifications

Core JavaScript, Core React.js (Coding Ninjas), Google Digital Garage.

Hobbies

Rural education NGO volunteer (3 years): assessed rural pupils' learning levels for government bodies and tutored on-site. | UTN Buddy Program mentor. | Marathon finisher 2025, 2026.